

Gestures and Words in Early Development of Children With Down Syndrome

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This study investigated the development of language and communication in children with Down syndrome (DS). More specifically, the aim was to examine the relations among verbal comprehension, verbal production, and gesture production in the very early stages of development. Forty children (age range: 10–49 months) with DS and 40 children with normal development (age range: 8–17 months) participated in this study. Children with DS came from two Italian health centers. The communicative and linguistic development of children with DS was measured by administering the Italian version of the MacArthur Communicative Development Inventory. The children with DS were severely delayed when compared with normally developing children in reaching the developmental stages. In such children a dissociation emerged between verbal comprehension and production, in favor of comprehension, whereas a synchronous development was found between vocal lexical comprehension and gestural production. The individual differences previously reported in these children are also evident in all domains examined. There were no significant differences between children with DS and typically developing controls matched for lexical comprehension on verbal production. However the two groups differed significantly in gestural development, suggesting a “gesture advantage” in children with DS compared with controls matched for word comprehension. Some possible reasons for this dissociative profile are discussed.

KEY WORDS: gestures, words, development, Down syndrome

Studies of language abilities in children with Down syndrome (DS) report general delays in language development, together with specific dissociations among different linguistic domains (i.e., comprehension vs. production) or subdomains (e.g., lexicon vs. morphosyntax) (Cardoso-Martins, Mervis, & Mervis, 1985; Caselli, Marchetti, & Vicari, 1994; Chapman, 1995; Fowler, 1990; Miller, 1992). These linguistic problems become increasingly evident as development progresses. For example, Fowler (1990) showed that, in the early stages, no differences are observed in the syntactic and semantic skills of children with DS and typically developing children of comparable mental age. However, older children with DS do show a deficit in morphosyntactic and phonological areas (i.e., below mental age), with lexical abilities relatively preserved (e.g., at mental age). Chapman (1995) noted that some specific aspects of morphosyntax are generally more delayed than we would expect given mean length of utterance (MLU). Rondal (1993) demonstrated that subjects with DS tend to use simple sentences in which function words (articles, prepositions, pronouns, etc.) are frequently omitted—a

tendency that persists for a long time. These results have recently been confirmed in Italian subjects. Comparing the language of adolescents with DS with those of typically developing children matched on MLU, Fabbretti, Pizzuto, Vicari, and Volterra (1997) have shown that the language abilities in production of the subjects with DS was characterized by deficits in some morphological aspects of language, with relatively sparing use of the lexical and syntactic domains. Although certain profiles may be defined as “typical,” it should be noted that there is a wide variation in the language skills observed among children with DS. As adults, some individuals never go beyond the production of single words or simple word combinations (equivalent to a linguistic age of 2–3 years), whereas other subjects may show considerably more sophisticated language skills (Rondal, 1995; Vallar & Papagno, 1993).

Many authors underscore that children with DS show lexical comprehension levels in keeping with their mental age through the first 5 years of life but sorely lag behind in terms of production. In some cases speech is unintelligible. First words are often comprehended and produced at an appropriate point given the child's mental age. However, subsequent growth in expressive vocabulary tends to be very slow, with a greater lag between lexical comprehension and lexical production than one observes in typically developing children (Beeghley & Cicchetti, 1986; Cardoso-Martins et al., 1985; Cicchetti & Beeghley, 1990).

In a detailed analysis of the language acquisition in children with DS, Miller (1988, 1992) emphasized the existence of a verbal production deficit. He noted that this deficit becomes increasingly evident after a mental age of 17 months. Two hypotheses have been developed in an attempt to explain the asynchrony between production and comprehension: a motor planning deficit and a memory deficit.

Miller (1992) has underscored problems in speech motor control in children with DS, neuromuscular deficits that impair the rapidly coordinated movements required by the speech production system (e.g., the lips, tongue, jaw, respiration, and phonation). In addition to these motor problems, frequent middle ear infections may impair the perception of sound, limiting the construction of an appropriate phonological representation and the ability to reproduce accurate speech sounds in production.

Several studies have reported a marked reduction of short-term memory capacity in persons with DS. Evidence includes reports of a shorter verbal span in subjects with DS than in persons with mental retardation of different etiology, matched for mental age (MA) (Hulme & Mackenzie, 1992; Marcell & Weeks, 1988; Vicari, Carlesimo, & Caltagirone, 1995). Hulme &

Mackenzie (1992) described an impaired articulatory loop in subjects with DS. There is evidence that the reduced functionality of the articulatory rehearsal mechanism plays a role in the learning of new phonological information and of new words (Hulme & Mackenzie, 1992). Furthermore, Pueschel (1988) suggested that, because of their short-term memory deficits, subjects with DS have difficulty in retaining information included in questions needed to respond correctly to a stimulus word.

In contrast to the specific deficits in verbal production, gestural communication in children with DS serves as an effective means of communication for a longer time than in typically developing children (Caselli, Longobardi, & Pisaneschi, 1997; Goodman, Wilson, & Bornstein, 1978; Mundy, Sigman, Kasari, & Yirmiya, 1989). However, most studies of gestural communication in children with DS have focused on the use of pointing. In particular, the research by Mundy et al. (1989) showed that, unlike children with autism, children with DS can produce pointing gestures with a declarative function, but they do have difficulty using pointing for an imperative-instrumental function. Franco and Wishart (1995) analyzed the production and comprehension of pointing gestures in a group of children with DS with a mean age of 37 months. Results showed that these children use pointing in an effective manner when interacting with their mother and with other children.

The relationship between gesture and the emergence of vocabulary in a single group of children with DS during the initial stages of language development remains unspecified. This is due, at least in part, to some substantial methodological problems involved in assessing early vocabulary. Because most children in the earliest stages of language development do not talk very much, the normal 30–60 minute recording session can grossly underestimate vocabulary size. Longer recording sessions may provide a better estimate, but they also present immense logistical problems for the family and for the experimenter. Parent reports have provided a viable alternative. There are several advantages to parent report for the assessment of early language development. First, it is cost-effective and time-efficient. Second, it takes advantage of the parents' extensive knowledge of their child's language abilities. Third, the child's performance is not affected by such factors as fatigue or lack of familiarity with the examiner (Fenson et al., 1993). However, because parent report may be subject to various forms of bias and lack of expertise, the establishment of instrument reliability and validity is essential.

One instrument that meets these criteria is the MacArthur Communicative Development Inventory (CDI), which records aspects of gesture, vocabulary, and

grammar when these aspects of language and communication first emerge. The CDI was originally developed in English (Fenson et al., 1993) and Italian (Caselli & Casadio, 1995), and it has proven to be highly reliable in assessing language abilities in typically developing children between 8 and 30 months of age. Furthermore, the validity of parental observations in English-speaking countries has been widely demonstrated (Dale, Bates, Reznick, & Morisset, 1989; Fenson et al., 1994; Rescorla, 1989; Reznick & Goldsmith, 1989).

In Italy, Camaioni, Caselli, Longobardi, and Volterra (1991) have demonstrated the reliability of answers provided by parents to a questionnaire on the communicative and linguistic development of children in their second year of life, compared with observations of the same children carried out by experts. Other studies have established the validity of the Italian version of the CDI: "Il Primo Vocabolario del Bambino" (PVB). Modena and Caselli (1996) have shown that lexical comprehension on the PVB Inventory correlates positively and significantly with a laboratory evaluation of lexical comprehension in a group of 30 children between 12 to 24 months of age. Devescovi, Caselli, and Bonanni (1996) reported that measures of linguistic production from the PVB Inventory correlate positively and significantly with results of a sentence repetition test administered to a group of 33 children between 24 and 30 months of age.

Although the value of parent report in typically developing children is well established, very few studies have investigated the reliability and validity of this instrument for the parents of children with disorders (e.g., developmental delays, language impairments). For children with DS, Miller and colleagues (Miller, 1992; Miller, Sedey, & Miolo 1995) have demonstrated significant correlations between the CDI parent report vocabulary checklist and measures of observed vocabulary in typically developing children and in children with DS. Furthermore, performance on the vocabulary checklist at approximately 20 months of mental age was highly correlated with later measurements of vocabulary development in the same group of children at approximately 28 months of mental age. The magnitude of the correlations reported and the similarity of the coefficients between the two subject groups support the validity of the CDI as a measure of vocabulary development in typically developing children and in young children with DS.

Stella, Lami, Caselli, Casadio, and Pizzoli (1993) have used the PVB Inventory to evaluate 18 Italian children with DS between the ages of 14 and 30 months. These subjects were divided into three subgroups with an average mental age of 7, 12, and 17 months, respectively. The aim of that study was to analyze the relationship between cognitive development and vocabulary

size in vocal comprehension, vocal production, and gesture production. Each subgroup was compared with a control group consisting of typically developing children whose chronological age corresponded to the mental age of the children with DS. The children with DS who had a mental age of 7 and 12 months had levels of communicative and linguistic development commensurate with their mental age. Some differences appeared, however, when the children with DS in the 17 month group were compared with control children. These differences included significantly lower vocal production and significantly higher gesture production. No difference was found in comprehension. These data are in accordance with those reported by Miller (1992), who points out the presence of a vocal production deficit after a mental age of 17 months.

The purpose of this study was to analyze the early stages of communicative and linguistic development in a wider sample of children with DS, comparing performance in the vocal and gestural modalities. The first aim was to measure the degree of communicative and linguistic deficit in children with DS as compared to their chronological age. This analysis allowed us to verify whether there are specific dissociations between vocal comprehension, vocal production, and gesture production in the first stages of development (including the emerging comprehension/production reported by Miller, 1996). The second aim was to learn more about the nature of these dissociations, by comparing children with DS and a control group of typically developing children matched for lexical comprehension. This procedure made it possible to assess differences between the two groups of children in terms of chronological age, word production, and gestural production. In fact, many studies report that there is a large discrepancy between the measured language abilities of children with DS and expectations based on mental age assessment (for a review see Chapman, 1995).

The present study will permit us to clarify whether the dissociation between comprehension and production, if it exists, involves only speech (significant delay in vocal production) or also gesture (significant delay in both vocal and gestural production). This perspective is important in addressing the issue of productive communication competencies in general as compared to those that are specifically vocal.

Method

Subjects

Forty children with DS participated in this study. These children, 62% boys and 38% girls, ranged in age from 10 to 49 months ($M = 28;3$). All children came from

two Italian health centers: one in Rome (“Bambino Gesù” Hospital, S. Marinella) and the other in Bologna (Center for Cognitive Disabilities). The sample was selected from a broader range of subjects using the following criteria: documented cytogenetic diagnosis of trisomy 21, absence of relevant ear-nose-throat problems, negative history of epilepsy or psychiatric disorders. The children, all in the holophrastic period, were not divided into subgroups by age. Social-cultural levels were defined in terms of the level of education and the jobs the parents held. In particular, we assigned a high level when parents had a university degree and were professionals, academics, or the like (8% of the families); a middle level when parents had graduated from high school and worked as preschool teachers, office workers, and so forth (63% of the families), and a low level when parents had attended only elementary, junior high, or a vocational school and were employed as blue-collar workers, clerks, and the like (29% of the families).

Instruments

The communicative and linguistic development of children with DS were measured by administering the Italian MacArthur questionnaire (Il Primo Vocabolario del Bambino, PVB; Caselli & Casadio, 1995) to the parents. There are two forms of the questionnaire: “Gestures and Words” and “Words and Phrases.” (The latter will not be described because it was not used in this study). The “Gestures and Words” form consists of a list of 408 words, divided into 19 categories that include nouns, verbs, and function words (e.g., articles, pronouns, prepositions) and a list of 63 gestures and actions, also grouped into categories (e.g., deictic and representational gestures, pretend actions, routines). For words, the parents had to mark those that the child understood and/or produced. As for gestures, parents were asked only whether the child produced a particular gesture or action.

The normative data (percentiles) referred to in this study are taken from the results of the “Gestures and Words” forms administered to 315 parents of Italian children between the ages of 8 and 17 months (Caselli & Casadio, 1995).

Procedure

The parents of the 40 children included in the sample were asked to fill in the questionnaire, and it was made clear that the objective was to collect detailed information about the children’s development. The questionnaire was left with the parents for about one week; they were asked to examine how it was organized and to begin filling it in only after observing their child for two or three days. Any questions or clarifications could

be asked of the person who would collect the completed questionnaire. All parents willingly participated in the data collection, showing interest in the instrument that made them observe their child more carefully and become more aware of his/her real abilities. All questionnaires were checked to make sure the parents had filled them in correctly and in full, then coded to determine the measurements of the verbal and gestural abilities of the children examined.

Results

Comprehension and Vocal/Gesture Production in Children With DS

Table 1 shows the descriptive analysis of the comprehension and production of words and gestures by the 40 subjects with DS. As is the case with typically developing children, average values in children with DS for word comprehension are notably greater than those for word production, revealing an asynchrony in the rate of lexical acquisition in these two domains. Marked individual differences between children emerge in the domains examined.

In order to evaluate the degree of retardation in communicative and linguistic abilities in children with DS, the values obtained for each subject in the three domains studied (verbal comprehension, verbal production, and gesture production) were compared with the normative scores (percentiles) reported in Caselli and Casadio (1995). Data on typically developing Italian children were collected using the same instrument: the “Gestures and Words” form of the PVB Inventory. This procedure made it possible to assign to each child with DS three “indexes” corresponding to a “Language/Communication Age” (in months) for word comprehension (CLA), for word production (PLA), and for gesture production (GCA).

Children with DS at an average chronological age (CA) of 28;5 months display an average language/communicative age of 14;2 months in word comprehension (CLA), 15;1 months in word production (PLA), and 14;6 months in gestural production (GCA). A MANOVA for repeated measures was performed to compare CA, CLA, PLA, and GCA. This analysis showed a general

Table 1. Mean number of words comprehended (WC), words produced (WP), and gestures produced (GP) by children with Down syndrome ($N = 40$).

	WC	WP	GP
<i>M</i>	166.4	26.5	37.4
<i>SD</i>	117.4	55.5	18.9
Range	0–390	0–302	2–62

Table 2. Down syndrome children's communicative development indices ($N = 40$).

Development indices	Paired differences						Sig. (2-tailed)	
	M	SD	Std. error Mean	95% Bonferroni confidence interval of the difference		t		df
				Lower	Upper			
CA-PLA	15.13	9.27	1.47	11.03	19.22	10.32	39	.000
CA-CLA	14.1	9.36	1.48	9.96	18.24	9.53	39	.000
CA-GCA	13.75	9.39	1.48	9.6	17.9	9.26	39	.000
PLA-GCA	-1.38	1.69	0.27	-2.12	-0.63	-5.15	39	.000
CLA-CCA	-0.35	1.35	0.21	-0.95	0.25	-1.64	39	.109
PLA-CLA	-1.02	1.95	0.31	-1.89	-0.16	-3.32	39	.002

Note. CA = chronological age, PLA = word production age, CLA = word comprehension age, GCA = gesture production age.

significant effect ($F = 88.29$; Greenhouse-Geisser corrected $df = 1.09, 41.55$; $p < .0001$). Post hoc tests (with a Bonferroni correction) were also used to explore statistically significant differences between pairs of measures. This analysis showed a general significant effect (Hotelling's $T^2 = 3.41$; $p < .001$). The results are summarized in Table 2.

As expected, there is a significant difference between chronological age and each of the three developmental age scores, confirming an overall delay that affects children with DS in the area of communicative and language abilities. When each of the three developmental indexes was compared directly, a significant difference emerged between CLA and PLA; there was also a significant difference between PLA and GCA, indicating a higher level of gestural abilities. No significant differences were found between CLA and GCA. These results correspond to findings for typically developing subjects (Caselli et al., 1995; Fenson et al., 1994), with vocal comprehension and gesture production developing ahead of vocal production.

Comparison of Children With DS and Typically Developing Children

Studies examining early communicative and language development in typically developing children have recently shown that certain important changes in this area are related not only to chronological age, but also to the developmental stage of the child in certain specific domains. For example, progress in vocal production and comprehension is often linked to a halt and gradual "drop" in the use of gestures; this changes the relationship between domains and between modalities (Iverson, Capirci, & Caselli, 1994; Volterra & Iverson 1995). Moreover, individual differences in the early stages of development are highly marked (Bates, Bretherton, & Snyder, 1988; Shore, 1995), sometimes

leading researchers to consider groups of subjects balanced, not according to chronological age, but according to internal linguistic measurements (in some cases comprehension, in others production) (Caselli & Casadio, 1995; Caselli et al., 1995; Fenson et al., 1994; Marchman & Bates, 1994). This type of approach proved to be particularly fruitful in this study given that there may be a discrepancy between the measured language abilities of children with DS and expectations based on mental age assessment (Chapman, 1995; Fowler, 1990).

Each subject with DS was matched to a typically developing subject with the same number of words comprehended, randomly selected from the normative sample ($N = 315$) of the validation study of the PVB Inventory. This produced a control sample of 40 typically developing children. Word comprehension was chosen as the basis for matching the two groups because, as already noted, the comprehension level is generally more consistent with the global cognitive level of children with DS (Miller, 1996; Stella et al., 1993). Table 3 shows the mean age of the control sample of typically developing children and the measurements (means and standard deviations) of their vocal and gestural production.

In order to assess the differences between children with DS and typically developing children in terms of

Table 3. Mean number of words produced (WP) and gestures produced (GP) by normally developing children ($N = 40$).

	CA (months)	WP	GP
<i>M</i>	13.6	20.4	28.6
<i>SD</i>	2.7	20.4	30
Range	8-17	0-111	0-113

Note. CA = chronological age.

chronological age, word production, and gestural production, a MANOVA was performed with group as the between-subjects factor and the three dependent variables as within-subject factors. The reason for such a procedure is the high correlation between the dependent variables (Mauchly's test of Sphericity, $p < .01$). The results showed that the two groups differed overall (Hotelling's $T^2 = 1.09$; $p < .001$) and in particular for chronological age (mean difference = 14.7, 95%; Bonferroni confidence interval = 10.2–19.2) and for gestural production (mean difference = 8.8; 95%; Bonferroni confidence interval = .4–17.2). No significant between-group differences were found for word production.

To learn more about these group differences in gesture production, percentages were calculated for each of the different types of gestures provided by the PVB Inventory (e.g., "pointing/showing/requesting," "symbolic communicative gestures," "games and routines," "imitating adults actions," etc.) out of the total number of gestures produced by each child in the two groups. In comparison with typically developing children at the same word comprehension level, the children with DS produced a greater percentage of symbolic communicative gestures such as "GONE," "HOT," "BE QUIET," "GOOD" (mean DS = 5.6; NC = 4.4), a greater percentage of "pretending gestures/actions"—for example, when playing with dolls and stuffed animals (mean DS = 5.9; NC = 3.6), and a greater percentage of "pretending gestures/actions," which involve an ability to perform symbolic transformations—for example, pretending that stones are food and a stick is a spoon (mean DS = 0.37; NC = 0.17).

To investigate these relationships in more detail, we divided both the sample of children with DS and the control sample into five subgroups in terms of different levels of verbal comprehension. The subgroups included children with a word comprehension repertoire of 0–50; 51–100; 101–200; 201–300; 301–400, respectively. The descriptive analyses for five levels of word comprehension are reported in Table 4.

Chronological age, as expected, is higher at higher levels of comprehension. The mean chronological age of children with DS was also much higher than that of typically developing children in all subgroups. A more interesting result lies in the fact that the lag between the two groups is wider in the higher levels of word comprehension. Moreover, important individual differences emerged within the group of children with DS. Some children with a chronological age of 34 months were at the first level of word comprehension (0–50 words), whereas some younger children (21 months) were at a more advanced level of lexical comprehension (201–300 words).

In line with the MANOVA results reported above, this analysis over comprehension levels did not reveal

Table 4. Word and gesture production in children with Down syndrome (DS) and normal development (ND) at different levels of word comprehension.

Subjects	Age	Words	Gestures
Word Comprehension: 0–50 (DS = 9; ND = 9)			
Down			
M	17.9	3.7	14.2
SD	6.9	5.5	10.7
Range	10–34	0–15	2–41
Normal			
M	10.3	2.2	13.1
SD	2.3	3.5	7.2
Range	8–16	0–11	5–26
Word Comprehension: 51–100 (DS = 7; ND = 7)			
Down			
M	21.4	2.4	21.3
SD	10.6	1.9	6.7
Range	14–45	0–6	11–29
Normal			
M	12.2	7.1	19.7
SD	1.9	8.2	12.2
Range	10–15	0–20	7–33
Word Comprehension: 101–200 (DS = 6; ND = 6)			
Down			
M	32.2	10.2	45.2
SD	9.4	7.1	11.8
Range	21–44	1–20	26–57
Normal			
M	13	11.1	30.5
SD	1.2	12	8
Range	12–15	0–27	18–41
Word Comprehension: 201–300 (DS = 10; ND = 10)			
Down			
M	33.5	23.5	46.9
SD	6.7	28.9	4.4
Range	21–44	4–103	37–52
Normal			
M	15.8	30	37.6
SD	0.87	37	6.4
Range	14–17	0–91	26–49
Word Comprehension: 301–400 (DS = 7; ND = 8)			
Down			
M	36.6	59	59.9
SD	9.7	55.7	1.7
Range	25–49	12–174	57–62
Normal			
M	16	47.1	41.1
SD	0.92	37.9	7.7
Range	14–17	0–111	24–50

important differences in the word production of typically developing children and children with DS at the same level of lexical comprehension. As for gesture production, there were no marked differences between typically developing children and children with DS within the first two subgroups. The gesture repertoire was, however, much richer in children with DS in the subgroups with higher word comprehension levels (from the level of 101–200 words upwards).

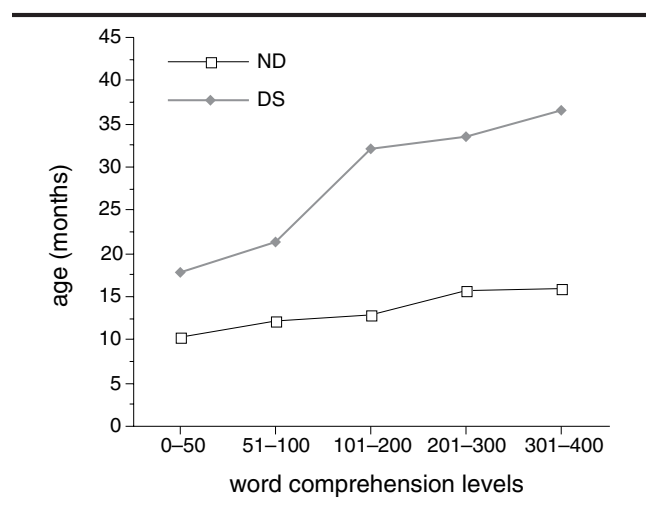
A MANOVA was performed with chronological age, verbal production, and gestural production as dependent variables, and group (two levels) and comprehension (five levels) as between factors. The aim of this analysis was to verify the presence of a group $\times$ age interaction and at which comprehension level the difference between children with DS and typically developing children reaches significance. For this statistical analysis, one outlier was excluded from the subgroup of children with DS at a word comprehension of 301–400 words, as established by a box-plot procedure (outliers $\geq$ median + 1.5 range interquartile).

The results showed that this interaction was significant for age and gestures (respectively, $F(4, 69) = 3.23, p < .02$ and $F(4, 69) = 3.51, p < .01$). By examining the contrast between levels, a significant increment was observed when word comprehension increased from 51–100 to 101–200 in the group of children with DS as compared to the controls (Figures 1, 2, and 3). No significant differences were found for word production.

Discussion

The purpose of this study was to analyze the early stages of communicative and linguistic development in

Figure 1. Chronological age of children with normal development (ND) and Down syndrome (DS) at different levels of word comprehension.



children with DS. In particular, we were interested in verifying the dissociation between word comprehension and production, matching children with DS with typically developing controls on the basis of a linguistic measure: lexical comprehension. We also analyzed the relationship between the vocal and gestural modalities to investigate whether the difficulties of children with DS in the domain of production are restricted to speech, or whether they involve the production of gestures as well.

The results of the present study confirm a severe delay in children with DS in all the areas of development considered here. In addition, as is often the case

Figure 2. Gesture production in children with normal development (ND) and Down syndrome (DS) at different levels of word comprehension.

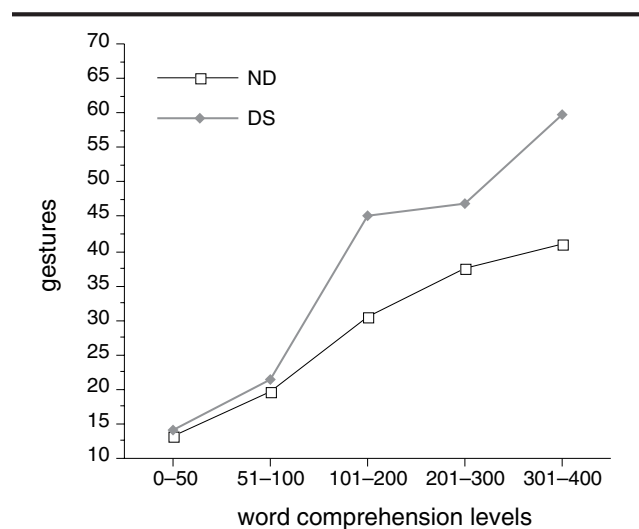
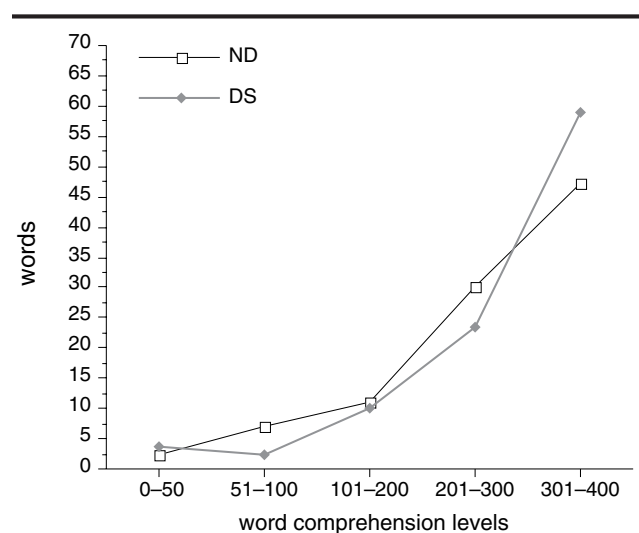


Figure 3. Word production in children with normal development (ND) and Down syndrome (DS) at different levels of word comprehension.



for typically developing children as well (Bates et al., 1988; Caselli & Casadio, 1995; Fenson et al., 1994), a dissociation emerged in children with DS between verbal comprehension and production, in favor of comprehension, whereas synchronous development was found in vocal lexical comprehension and gestural production. Substantial individual differences were observed in all domains examined.

In comparing children with DS and controls matched on lexical comprehension levels, no significant differences between the two groups were found for vocal production. The lag between comprehension and production appears to be quite similar for children with DS and typically developing controls when the two groups are matched for word comprehension level. The two groups did differ, however, in gestural production. Comparison with typically developing children revealed a significant difference in favor of the group of children with DS, although this difference only becomes apparent when lexical comprehension exceeds 100 words.

A detailed look at the specific types of gestures involved in this group difference suggests that children with DS produce a greater percentage of gestures in categories that imply more advanced cognitive skills, including symbolic communicative gestures, pretending gestures, and actions which involve an ability to perform symbolic transformations. These results pose an interesting problem with regard to the relationship between language and cognition and the relationship between gesture and word production. Many studies focused on early development in typically developing children have led to the conclusion that the same symbolic processes control the production of first words, first symbolic/referential gestures, and the ability to produce pretend actions. These abilities develop in parallel in both the gestural and vocal modalities (Bates, Benigni, Bretherton, Camaioni, & Volterra, 1979; McCune-Nicolich, 1981; Piaget & Inhelder, 1986). Moreover, some studies describe three developmental stages in gestural and vocal production. In the first period, children use more gestures, particularly deictics, than words. In this period, words and gestures are essentially presymbolic-communicative signals. In the second period, children produce the same number of gestures and words, which progressively achieve a symbolic/representational status. Some authors define this as “the period of the equipotentiality” of gestures and words. In the third period, the number of words used by children increases significantly while the number of symbolic/referential gestures progressively drops (Iverson et al., 1994; Volterra & Caselli, 1986; Volterra & Iverson, 1995).

The children with DS studied in the present research seem to follow a different pattern. During the initial stages, their gestural and vocal production are similar

to those of typically developing children matched for word comprehension. Subsequently, however, symbolic communicative gestures and actions continue to increase and appear to be more sophisticated in children with DS than might be expected for their level of word comprehension and/or word production. Our data agree with those reported by other authors (Franco & Wishart, 1995; Singer Harris, Bellugi, Bates, Jones, & Rossen, 1997), who have demonstrated that the greater ability shown by children with DS in using communicative gestures results from their longer social experience (because of greater chronological age). This, in turn, leads to a sort of “specialization” in nonverbal communication.

The results of this study seem to support neither the hypothesis of a general motor production deficit nor that of a specific speech delay, at least at the levels of development considered. In particular, the failure to confirm a specific retardation in vocal lexical production (often reported in the literature; see Chapman, 1995 for a recent review) suggests that two factors should be considered: (1) the stage of development examined, and (2) the methodology used in collecting and analyzing data.

With regard to the first point, the level of word production that we have observed in children with DS corresponds to the word production level of typically developing 13-month-olds matched for comprehension. At that age, word production is an emerging ability, even in typically developing children, with strong individual variability; these two facts may mask a difference between the two groups that can only emerge when both groups move past the point of first words. In that regard, our data are consistent with those reported in other studies using the same instrument (Miller, 1992, 1996; Stella et al., 1993). In those studies, significant differences in lexical production emerged between children with DS and typically developing children matched on MA, but this difference was not evident before a mental age of 17 months.

This brings up a final issue: the specificity of the developmental profile in DS compared with other forms of mental retardation. Our results are consistent with a recent study by Singer Harris et al. (1997), who compared development of communication and language in children with DS with that of a group of children with Williams syndrome (WS). Data were collected by administering the MacArthur CDI to parents. In accord with the data presented here, Singer et al. found a significant advantage in gestural communication in children with DS compared with a sample of typically developing children at the same word production level and with a group of children with WS who did not differ from the DS group in chronological age. However, Singer Harris et al. did not find evidence for a difference between children

with DS and children with WS on either word comprehension or word production during the very early stages of development. This result was surprising, in view of reports that older children with WS are more advanced in verbal production than children with DS. However, Singer Harris et al. did find a significant group difference in expressive language among their older subjects, with higher levels of grammatical production in children with WS. In line with our interpretations in the present study, the authors concluded that group differences do not emerge until children move beyond a mental age (and a language age) roughly corresponding to 17–18 months in children who are developing on a typical schedule.

This study confirms the utility of the PVB Inventory with parents of children with DS. However, such parent report measures yield a purely quantitative estimate of the types of words and gestures produced by children. Analysis of the use of communicative gestures and words by children with DS may reveal a different view of their abilities than that obtained in the present study. Indeed, the results of an observational study on the relationship between early words and communicative gestures and the frequency of use of the two modalities in three children with DS showed a marked preference for gestural communication. Even when the word repertoire was numerically similar or greater than that of gestures, the latter remained the children's preferred form of communication (Caselli et al., 1997). It seems fair to conclude from our own results and many other studies that gestural communication is an alternative form of communication for children with DS, one that they may use to compensate for the problems in speech production that emerge by a mental age of approximately 17 months.

Finally, these results are compatible with an important shift in perspective regarding the relationship between language and cognition in children with mental retardation. Children with DS display a profile of language and communicative development characterized by a specific delay in verbal production and a significant preference for gestural communication (Chapman, 1995; Fabbretti et al., 1997; Fowler, 1990; Miller, 1988, 1992). The same pattern has not been reported in other forms of mental retardation. The most relevant comparisons are those involving children with WS, who present with a markedly different profile from the one reported for children with DS. For example, some authors have argued that children with WS are spared in expressive language for their mental age, despite cognitive delays equivalent to those reported for children with DS (Bellugi, Marks, Bihrlé, & Sabo, 1988; Karmiloff-Smith, 1992). There is some controversy regarding the nature of the profile in WS, due in part to differences in the age and developmental level. For example, studies conducted

with adolescents have described proficient performances in both comprehension and production (Bellugi, Bihrlé, Jernigan, Trauner, & Doherty, 1990; Bellugi, Wang, & Jernigan, 1993). Other reports concerning younger children with WS have shown impaired linguistic competence (Crisco, Dobbs, & Mulhern, 1988; Gosh, Stading, & Pankau, 1994; Udwin & Yule, 1990). Despite these variations, the specific profile of deficits differs from the profile reported for children with DS, particularly with regard to the difference between modalities that we have studied here (e.g., a gestural advantage in DS that is not observed in WS (Singer Harris et al., 1997). In fact, some authors (Mervis & Bertrand, 1993) have reported that young children with WS produce referential speech well before beginning to produce communicative gestures. In almost every respect, this profile differs from the one that has been reported for children with DS, where any difference in modality that does emerge tends to favor comprehension more than production and gesture more than speech.

Further contrasts between people with DS and WS emerge from a comparison of within-language dissociations. For example, Rubba and Klima (1991), in describing the language production of subjects with WS, recognized frequent errors in preposition use. Studies conducted in languages with a more complex morphology and syntax than English, such as Italian, have documented problems for children with WS in category fluency, naming, and comprehension of complex morphosyntactic sentences (Volterra, Capirci, Pezzini, Sabbadini, & Vicari, 1996). Volterra et al. (1996), who reported impaired semantic fluency and naming but adequate comprehension of single words in children with WS, suggested a deficit in these subjects in deliberately accessing their lexical repertoire. In sum, the linguistic profile of subjects with WS appears to be characterized by “peaks and valleys”; and some aspects, such as phonological fluency, seem to be more efficient than others, such as grammar and semantics. This contrasts with the specific profile of sparing and impairment that has been reported for subjects with DS, including greater impairments overall in expressive language, with marked deficits in specific aspects of phonology and morphosyntax.

In conclusion, mental retardation is not a uniform condition, characterized by an undifferentiated delay of cognitive development. Instead, mental retardation can be characterized in terms of one or more deficits within a complex cognitive system in which some cognitive abilities can be disrupted more than others (Detterman, 1987; Vicari et al., 1995). A neuropsychological approach to the study of cognitive and language abilities may provide, in this sense, invaluable information about the qualitative features of the intellectual deficit in relation to the typically developing population and to groups with mental retardation of different etiologies.

Acknowledgments

Partial support for this project during the period of manuscript preparation was provided by the grant from the John D. and Catherine T. MacArthur Foundation Research Network on Early Childhood Transitions and by NIN/NIDCD PO50 "Origins of Communicative Disorders" grant to Elizabeth Bates. The authors wish to thank Elizabeth Bates, Virginia Volterra, and Paola Casadio for their suggestions and criticism on a preliminary version of this work. We would like also to thank Patrizio Pasqualetti for help in data analysis and two anonymous referees for their helpful comments on an earlier version of the manuscript.

References

- Bates, E., Benigni, L., Bretherton, I., Camaioni, L., & Volterra, V. (Eds.). (1979). *The emergence of symbols: Cognition and communication in infancy*. New York: Academic Press.
- Bates, E., Bretherton, I., & Snyder, L. (1988). *From first words to grammar: Individual differences and dissociable mechanisms*. New York: Cambridge University Press.
- Beeghly, M., & Cicchetti, D. (1986). *Early language development in children with Down syndrome: A longitudinal study*. Paper presented at the 11th Annual Boston University Conference on Language Development, Boston.
- Bellugi, U., Marks, S., Bihrl, A. M., & Sabo, H. (1988). Dissociation between language and cognitive functions in Williams syndrome. In D. Bishop & K. Mogford (Eds.), *Language development in exceptional circumstances* (pp. 177–189). London: Churchill Livingstone.
- Bellugi, U., Bihrl, A. M., Jernigan, T. L., Trauner, D., & Doherty, S. (1990). Neuropsychological, neurological and neuroanatomic profile of Williams syndrome. *American Journal of Medical Genetics*, 6, 115–125.
- Bellugi, U., Wang, P. P., & Jernigan, T. L. (1993). Williams syndrome: An unusual neuropsychological profile. In S. H. Broman & J. Grafman (Eds.), *Atypical cognitive deficit in developmental disorders* (pp. 23–56). Hillsdale, NY: LEA.
- Camaioni, L., Caselli, M. C., Longobardi, E., & Volterra, V. (1991). A parent report instrument for early language assessment. *First Language*, 11, 345–359.
- Cardoso-Martins, C., Mervis, B., & Mervis, C. (1985). Early vocabulary acquisition by children with Down syndrome. *American Journal of Mental Deficiency*, 90, 177–184.
- Caselli, M. C., & Casadio, P. (1995). *Il primo vocabolario del bambino*. Milano: Franco Angeli.
- Caselli, M. C., Bates, E., Casadio, P., Fenson, J., Fenson, L., Sanderl, L., & Weir, J. (1995). A cross-linguistic study of early lexical development. *Cognitive Development*, 10, 159–199.
- Caselli, M. C., Longobardi, E., & Pisaneschi, R. (1997). Gesti e parole in bambini con sindrome di Down. *Psicologia Clinica dello Sviluppo*, 1, 45–63.
- Caselli, M. C., Marchetti, M. C., & Vicari, S. (1994). Conoscenze lessicali e primo sviluppo morfosintattico. In A. Contardi & S. Vicari (Eds.), *Le persone Down* (pp. 28–48). Milano: Franco Angeli.
- Chapman, R. S. (1995). Language development in children and adolescents with Down syndrome. In P. Fletcher & B. MacWhinney (Eds.), *The handbook of child language* (pp. 641–663). Oxford: Blackwell.
- Cicchetti, D., & Beeghly, M. (Eds.). (1990). *Children with Down syndrome. A developmental perspective*. Cambridge, U.K.: Cambridge University Press.
- Crisco, J., Dobbs, J., & Mulhern, R. (1988). Cognitive processing of children with Williams syndrome. *Developmental Medicine and Child Neurology*, 30, 650–656.
- Dale, P. S., Bates, E., Pethick, S., & Morisset, C. (1989). The validity of a parent report instrument of child language at 20 months. *Journal of Child Language*, 16, 239–250.
- Devescovi, A., Caselli, M. C., & Bonanni, R. (1996, October). *Valutazione del primo sviluppo morfosintattico: Confronto fra due strumenti di rilevazione*. Paper presented to X Congress of Italian Association of Psychology—Section of Developmental Psychology. Capri, Italy.
- Detterman, D. K. (1987). Theoretical notions of intelligence and mental retardation. *American Journal of Mental Deficiency*, 92, 2–11.
- Fabbretti, D., Pizzuto, E., Vicari, S., & Volterra, V. (1997). A story description task in children with Down syndrome: Lexical and morphosyntactic abilities. *Journal of Intellectual Disability Research*, 41, 165–179.
- Fenson, L., Dale, P. S., Reznick, J. S., Thal, D., Bates, E., Hartung, J. P., Pethick, S., & Reilly, J. S. (1993). *The MacArthur Communicative Development Inventories: User's guide and technical manual*. San Diego: Singular Publishing Group.
- Fenson, L., Dale, P. S., Reznick, J. S., Bates, E., Thal, D., & Pethick, S. (1994). Variability in early communicative development. *Monograph of the Society for Research in Child Development*, 59 (5), Serial No. 242.
- Fowler, A. E. (1990). Language abilities in children with Down syndrome: Evidence for a specific syntactic delay. In D. Cicchetti & M. Beeghly (Eds.), *Children with Down syndrome: A developmental perspective* (pp. 302–328). Cambridge, U.K.: Cambridge University Press.
- Franco, F., & Wishart, J. (1995). The use of pointing and other gestures by young children with Down syndrome. *American Journal of Mental Retardation*, 100(2), 160–182.
- Goodman, L., Wilson, P., & Bornstein, H. (1978). Results of a national survey of sign language programs in special education. *Mental Retardation*, 16, 104–106.
- Gosh, A., Stading, G., & Pankau, R. (1994). Linguistic abilities in children with Williams-Beuren syndrome. *American Journal of Medical Genetics*, 52, 291–296.
- Hulme, C., & Mackenzie, S. (1992). *Working memory and severe learning difficulties*. Hove: LEA.
- Iverson, J., Capirci, O., & Caselli, M. C. (1994). From communication to language in two modalities. *Cognitive Development*, 9, 23–43.
- Karmiloff-Smith, A. (1992). *Abnormal phenotypes and challenges they pose to connectionist models of development* (Tech. Rep. PDP. CNS. 92.7). London: Medical Research Council, Cognitive Development Unit.
- Marcell, M. M., & Week, S. L. (1988). Short-term memory

- difficulties and Down syndrome. *Journal of Mental Deficiency Research*, 32, 153–162.
- Marchman, V., & Bates, E.** (1994). Continuity in lexical and morphological development: A test of the critical mass hypothesis. *Journal of Child Language*, 21, 339–366.
- McCune-Nicolich, L.** (1981). Toward symbolic functioning: Structure of early pretend games and potential parallels with language. *Child Development*, 52, 785–797.
- Mervis, C. B., & Bertrand, J.** (1993, March). *General and specific relations between early language and early cognitive development*. Paper presented at the biennial meeting of the Society for Research in Child Development. New Orleans, LA.
- Miller, J. F.** (1996, October). *Early vocabulary acquisition in children with Down syndrome*. Paper presented at the International Congress on People with Down Syndrome Toward 2000. Rome, Italy.
- Miller, J. F.** (1988). The developmental asynchrony of language development in children with Down syndrome. In L. Nadel (Ed.), *Psychobiology of Down syndrome* (pp. 167–198). Cambridge, MA: MIT Press.
- Miller, J. F.** (1992). Development of speech and language in children with Down syndrome. In J. Y. Lott & E. E. McCoy (Eds.), *Clinical care for persons with Down syndrome* (pp. 39–50). New York: Academic Press.
- Miller, J. F., Sedey, A. L., & Miolo, G.** (1995). Validity of parent report measures of vocabulary development for children with Down syndrome. *Journal of Speech and Hearing Research*, 38, 1037–1044.
- Modena, I., & Caselli, M. C.** (1996). Valutare la comprensione del linguaggio nel secondo anno di vita: Sperimentazione di un test e utilizzazione di un questionario. *Rassegna di Psicologia*, 3, 71–90.
- Mundy, P., Sigman, M., Kasary, C., & Yirmiya, N.** (1989). Nonverbal communication skills in Down syndrome children. *Child Development*, 59, 235–249.
- Piaget, J., & Inhelder, B.** (1986). *La psicologia de l'enfant*. Paris: Presses Universitaires de France.
- Pueschel, S. M.** (1988). Visual and auditory processing in children with Down syndrome. In L. Nadel (Ed.), *The psychobiology of Down syndrome*. Cambridge, MA: MIT Press.
- Rescorla, L.** (1989). The language development survey: A screening tool for delayed language in toddlers. *Journal of Speech and Hearing Disorders*, 54, 587–599.
- Reznick, J. S., & Goldsmith, L.** (1989). A multiple form word production checklist for assessing early language. *Journal of Child Language*, 1, 91–100.
- Rondal, J. A.** (1993). Down's syndrome. In D. Bishop & K. Mogford (Eds.), *Language development in exceptional circumstances*. Hove: LEA.
- Rondal, J. A.** (1995). *Exceptional language development in Down syndrome: Implication for the cognition-language relationship*. New York: Cambridge University Press.
- Rubba, J., & Klima, E. S.** (1991). Preposition use in a speaker with Williams syndrome: Some cognitive grammar proposals. *Center for Research in Language Newsletter*, 3, 367–391.
- Singer Harris, N., Bellugi, U., Bates, E., Jones, W., & Rossen, M.** (1997). Contrasting profiles of language development in children with Williams and Down syndrome. *Developmental Neuropsychology*, 13, 345–370.
- Shore, C.** (1995). *Pathways into language: Individual differences in language development*. Thousand, Oaks, CA: Sage Publications.
- Stella, G., Lami, L., Caselli, M. C., Casadio, P., & Pizzoli, C.** (1993, July). *Evaluation of language development of children with Down syndrome in early childhood*. Poster presented at the Sixth International Congress for the Study of Child Language. Trieste: Italy.
- Vallar, G., & Papagno, C.** (1993). Preserved vocabulary acquisition in Down's syndrome: The role of phonological short-term memory. *Cortex*, 29, 467–483.
- Vicari, S., Carlesimo, A., & Caltagirone, C.** (1995). Short-term memory in persons with intellectual disabilities and Down syndrome. *Journal of Intellectual Disabilities Research*, 39, 532–537.
- Udwin, O., & Yule, W.** (1990). Expressive language of children with Williams syndrome. *American Journal of Medical Genetics*, 6, 108–114.
- Volterra, V., & Iverson, J.** (1995). When do modality factors affect the course of language acquisition? In K. Emmory & J. Rely (Eds.), *Language, gesture, and space* (pp. 371–390). Hillsdale, NJ: LEA.
- Volterra, V., Capirci, O., Pezzini, G., Sabbadini, L., & Vicari, S.** (1996). Linguistic abilities in Italian children with Williams syndrome. *Cortex*, 32, 663–677.
- Volterra, V., & Caselli, M. C.** (1986). First stage of language acquisition through two modalities in deaf and hearing children. *Italian Journal of Neurological Sciences*, 5, 109–115.

Received January 21, 1997

Accepted January 21, 1998

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